

Design and Analysis of Algorithm (CS3004-1)

Multiple Choice Questions

Unit-1

- 1` Main objective of analysing the algorithm is
 - a) To design algorithms
 - b)To implement algorithms
 - c) To evaluate for efficiency**
 - d) To debug algorithms
- 2 Which of the following is NOT a characteristic of a good algorithm?
 - a) Clarity and simplicity
 - b) Efficiency in terms of time and space
 - c) Correctness
 - d) The use of complex data structures**
- 3 What does "optimization" refer to in the context of algorithms?
 - a) The process of making an algorithm more complex
 - b) The process of improving an algorithm's efficiency**
 - c) The process of making an algorithm run slower
 - d) The process of making an algorithm more clear
- 4 What is the purpose of pseudocode in algorithm development?
 - a) To provide a step-by-step implementation guide**
 - b) To prevent others from understanding the
 - c) To hide the algorithm's logic
 - d) To succinct the algorithm
- 5 Characteristics of an algorithm are
 - a) Input, output, definite, finite, connective.
 - b) input, output, definite, finite, effective.**
 - c) input, output, reflective, finite, effective
 - d) input, output, reflective, finite, connective.
- 6 When an algorithm is written in the form of a programming language, it becomes a _
 - a) Flowchart
 - b) Program
 - c) Pseudo code**
 - d) Syntax
- 7 Pseudocode is written using
 - a)Combination of Native & programming language**
 - b) High level Programming language only
 - c) Natural language only
 - d) Low level programming language
- 8 Which of the following is incorrect?
Algorithms can be represented:
 - a)as pseudo codes
 - b)as syntax**
 - c) as programs
 - d) as flowcharts
- 9 What will be the output of the following pseudocode if the driver is fun(5)?
function fun(n)
 if n == 1 return 1 else return n + fun(n-1)
 - a)120
 - b)24
 - c)6
 - d)15**
- 10 What will be the output of the following pseudocode if the driver is fun(18,24)
function fun(a,b)
 while(b!=0)
 fun(b, a mod b)

b)24

c) 6

d)2

Given

ich d

11` Given the following time complexities, which one is the slowest growing function as $n \rightarrow \infty$

a) $O(\log n)$

b) $O(n)$

c) $O(n^2)$

d) $0(2^n)$

12 Which of the following best describes the Big- Ω notation?

a) Describes the lower bound of the time complexity of an algorithm

- b) Describes the upper bound of the time complexity of an algorithm

c) Describes the exact time complexity of an algorithm

d) Describes the average-case time complexity of an algorithm

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What does the Big-O notation, $O(f(n))$, represent?

a) The average-case time complexity of an algorithm

b) The worst-case time complexity of an algorithm

c) The best-case time complexity of an algorithm

d) The exact time taken by the algorithm

14 The time complexity of an algorithm is $O(n^2)$

Which of the following statements is correct about this algorithm?

a) The algorithm's running time grows linearly with respect to the size of the input

b) The algorithm's running time grows quadratically with respect to the size of the input

c) The algorithm's running time grows logarithmically with respect to the size of the input

d) The algorithm's running time is constant irrespective of the input size

15 If an algorithm's time complexity is $O(2^n)$, what can you infer about the algorithm?

a) The algorithm will be efficient for large inputs

b) The algorithm's time complexity grows exponentially with the input size

c) The algorithm will run in constant time for all inputs

d) The algorithm's runtime will decrease as the input size increases

16 In asymptotic analysis, what does it mean if a function is said to be $O(n)$?

a) The time complexity increases exponentially as n increases

b) The algorithm's runtime grows linearly with the input size

c) The runtime of the algorithm is independent of the input size

d) The algorithm's runtime grows logarithmically as n increases

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If an algorithm has a worst-case time complexity of $O(n^3)$, which of the following time complexities would indicate a faster algorithm for large n ?

a) $O(n^2)$

b) $O(n^3)$

c) $O(n^4)$

d) $O(2^n)$

18 Consider an algorithm with the following time complexities for different scenarios:

- Best Case: $O(n)$
- Worst Case: $O(n^2)$

- In which case will the algorithm perform the fastest?

 - Best Case**
 - Worst Case
 - Average Case
 - All cases will perform equally fast

19 Which of the following is the correct asymptotic notation for expressing the upper bound of an algorithm's running time?

 - $\Omega(n)$
 - $\Theta(n)$
 - $O(n)$**
 - $o(n)$

20 If an algorithm has a time complexity of $O(n^3)$, which of the following functions represents its growth rate?

 - Linear
 - Cubic**
 - Exponential
 - Logarithmic

21 What is the first step in the general plan for analyzing the time efficiency of recursive algorithms?

 - Identify the basic operation
 - Decide on a parameter indicating input size**
 - Set up a recurrence relation
 - Check whether the number of times the basic operation is executed can vary on different inputs

22 Which is the fundamental operation that is considered while analysing the algorithms?

 - Count of basic operation**
 - Arithmetic operation
 - Recurrence Equation
 - Geometric operation

23 What is the goal of the Tower of Hanoi problem?

 - To move all the disks to the second peg
 - To move all the disks to the first peg
 - To move all the disks to the third peg**
 - To move all the disks to the fourth peg

24 The time complexity of the solution tower of hanoi problem using recursion is

 - $O(n^2)$
 - $O(2^n)$**
 - $O(n \log n)$
 - $O(n)$

25 What is the typical time complexity of a non-recursive algorithm that processes each element in an array of size n exactly once?

 - $O(1)$
 - $O(n \log n)$
 - $O(n)$**
 - $O(n^2)$

26 Which of the following statements about brute force algorithms is true?

 - They always provide the optimal solution but are inefficient for large problem sizes.
 - They are the most efficient algorithms for all types of problems
 - They exhaustively search all possibilities, making them simple but often inefficient for large inputs**
 - They reduce the problem size in each iteration, unlike recursive algorithms

27 What is the primary characteristic of a brute force algorithm?

- a) It uses a divide-and-conquer strategy to break problems into smaller subproblems. b) **It attempts to solve the problem by checking all possible solutions until the correct one is found.**
- c) It uses dynamic programming to avoid redundant calculations d) It uses a greedy strategy to find an optimal solution.
- 28 What is the number of times the comparison operation is executed in the algorithm for finding the largest element in a list of numbers?
- a) $C(n) = n$ b) $C(n) = n - 1$
c) $C(n) = n + 1$ d) **$C(n) = n - 1$**
- 29 What is considered as the basic operation in the algorithm for finding the largest element in an array of integers?
- a) The assignment operation b) **The comparison operation**
c) The addition operation d) The multiplication operation
- 30 Which of the following recursive algorithms has a time complexity of $O(n)$?
- a) Fibonacci sequence using naive recursion. b) Merge sort.
c) Binary search on a sorted array. d) **Factorial computation using recursion.**
- 31 The Selection Sort algorithm finds the lowest value in an array and moves it to the ____ of the array.
- a) end b) **front**
c) middle d) particular location
- 32 The Bubble Sort algorithm finds the lowest value in an array and moves it to the ____ of the array.
- a) **end** b) front
c) middle d) particular location
- 33 Linear search algorithm, also known as ____
- a) Binary search b) simple search
c) **sequential search** d) Difficult search
- 34 **Exhaustive Search algorithm checks every possible solution to find the ____ one**
- a) Worst b) Average
c) amortized d) **Best**
- 35 **Exhaustive search falls under which algorithm technique**
- a) **Brute force** b) Decrease and Conquer
c) Divide and Conquer d) Greedy
- 36 A string-searching algorithm, sometimes called string-matching algorithm, is an algorithm that searches a body of text for portions that match by ____.
- a) **pattern** b) word
c) index d) text
- 37 The time complexity of the naive string matching algorithm $-m+1$
- a) $O(nm)$ b) $O(n+m)$
c) **$O(n-m+1)$** d) $O(n)$
- 38 The time complexity of the linear Search algorithm
- a) $O(nm)$ b) $O(n+m)$

- c) $O(n-m+1)$ d) $O(n)$

39 The time complexity of the Bubble sort algorithm
 a) $O(nm)$ b) $O(n^2)$
 c) $O(n-m+1)$ d) $O(n)$

40 The time complexity of the Selection sort algorithm
 a) $O(nm)$ b) $O(n^2)$
 c) $O(n-m+1)$ d) $O(n)$

41 What is the primary characteristic of the Divide and Conquer approach?
 a) Solving problems by iteration b) Solving problems by dividing them into overlapping sub problems
 c) **Dividing a problem into smaller independent subproblems, solving them recursively, and combining their results** d) Dividing a problem into smaller problems and linearly solving it.

42 In Quick Sort, what is the worst-case time complexity?
 a) $O(n^2)$ b) $O(n \log n)$
 c) $O(\log n)$ d) $O(n)$

43 Binary Search works only on which type of arrays?
 a) Unsorted arrays b) **Sorted arrays**
 c) Arrays with distinct elements d) Arrays of integers

44 Which of the following algorithms is **NOT** based on the Divide and Conquer approach?
 a) Merge Sort b) Quick Sort
 c) Binary Search d) **Selection sort**

45 Which of the following is true about Merge Sort?
 a) It is an in-place sorting algorithm b) It has a worst-case time complexity of $O(n^2)$
 c) **It uses additional memory for merging subarrays** d) It always selects the first element as a pivot

46 During the merging process of Merge Sort, what is the time complexity to merge two sorted subarrays?
 a) $O(n)$ b) $O(\log n)$
 c) $O(n^2)$ d) $O(n \log n)$

47 What is the average-case time complexity of merge sort?
 a) $\Theta(\log n)$ b) $\Theta(n)$
 c) $\Theta(n^2)$ d) **$\Theta(n \log n)$**

48 How many sub arrays does the quick sort algorithm divide the entire array into?
 a) one b) **two**
 c) three d) four

49 What is the recurrence relation for merge sort algorithm?
 a) $C(n)=2C(n)+1$ b) $C(n)=2C(n/2)-1$
 c) **$C(n)=2C(n/2)+n$** d) $C(n)=2C(n^2/2)+n$

50 **What is the time complexity of Binary Search in the best case?**
 a) $\Theta(n)$ b) $\Theta(\log n)$

c) $\Theta(n \log n)$

d) $\Theta(1)$

Design and Analysis of Algorithm (CS3004-1)

Multiple Choice Questions

Unit- 2

- 1` What is the primary idea behind the "Decrease and Conquer" technique?
a) Increase the problem size incrementally b) Solve the entire problem in one step
c) **Reduce the problem size at each step and solve it recursively** d) Divide the problem into multiple subproblems of equal size
- 2 Which of the following is a characteristic of the "Decrease and Conquer" method?
a) It always involves dividing the problem into two equal parts b) **It solves the problem by repeatedly reducing it to a smaller subproblem**
c) It uses dynamic programming to store intermediate results d) It is only applicable to sorting problems
- 3 In Insertion Sort, the problem size is reduced by:
a) Splitting the array into two parts of equal size b) Sorting only the largest element in each iteration
c) **Sorting a single element at a time** d) Exchanging adjacent elements
- 4 Which algorithm uses a variable-size decrease approach?
a) Exponentiation by squaring b) Binary search
c) **Euclid's algorithm** d) Insertion sort
- 5 Binary Search is a classic example of which variation of Decrease and Conquer?
a) Decrease by a constant b) **Decrease by a constant factor**
c) Decrease by one d) Divide and conquer
- 6 What type of data structure is used in Depth-First Search (DFS)?
a) Queue b) **Stack**
c) Priority Queue d) Heap
- 7 DFS is more suitable than BFS for which of the following?
a) Finding the shortest path in an unweighted graph b) Exploring all vertices in a level-by-level manner
c) **Detecting cycles in a graph** d) Solving shortest path problems in weighted graphs
- 8 What is the key difference between BFS and DFS in their approach?
a) BFS uses a stack; DFS uses a queue b) **BFS explores neighbors level by level; DFS explores as deep as possible along one branch**
c) BFS works only on trees; DFS works only on graphs d) BFS is recursive; DFS is iterative
- 9 Which of the following is true for BFS?

- a) It always uses recursion
- b) **It can handle graphs with cycles by avoiding revisits using a "visited" array**
- c) It cannot be used to check graph connectivity
- d) It is suitable only for weighted graphs
- 10 Which of the following problems can be solved using DFS?
- a) Finding connected components in a graph
- b) Finding the shortest path in an unweighted graph
- c) Topological sorting of a directed acyclic graph (DAG)
- d) **Both a and c**
- 11 **What is the primary application of topological sorting?**
- a) Finding the shortest path in a graph
- b) Sorting a list of elements in ascending order
- c) **Ordering the vertices of a directed acyclic graph (DAG) such that for every edge (u,v), vertex u comes before vertex v**
- d) Finding the minimum spanning tree of a graph
- 12 **Which of the following is a necessary condition for a graph to have a topological sorting?**
- a) The graph must be a tree
- b) **The graph must be a DAG**
- c) The graph must be a complete graph
- d) The graph must be a bipartite graph
- 13 **What is the time complexity of the topological sorting algorithm using Kahn's algorithm?**
- a) **$O(V + E)$**
- b) $O(V * E)$
- c) $O(E^2)$
- d) $O(V^2)$
- 14 **Which of the following is a correct statement about topological sorting?**
- a) Topological sorting is only possible for graphs with a single source vertex
- b) Topological sorting is only possible for graphs with a single sink vertex
- c) **Topological sorting can be performed on any DAG**
- d) Topological sorting can only be performed on graphs with a single cycle
- 15 **What happens if a graph has a cycle and we try to perform topological sorting on it?**
- a) The algorithm will produce a correct topological sorting
- b) The algorithm will produce an incorrect topological sorting
- c) **The algorithm will detect the cycle and report an error**
- d) The algorithm will go into an infinite loop
- 16 **Which of the following is an example of the Transform and Conquer method?**
- a) Merge Sort
- b) Binary search
- c) matrix multiplication
- d) **Heap sort**
- 17 **What operation is performed on an AVL tree when the balance factor of a node becomes greater than 1 or less than -1?**
- a) **Rotation**
- b) Deletion
- c) Insertion
- d) Search
- 18 **What is the condition for a binary search tree to be considered an AVL tree?**

- a) The height of the left subtree and the right subtree of every node differs by at most 2
- c) The height of the left subtree and the right subtree of every node is always equal
- b) The height of the left subtree and the right subtree of every node differs by at most 1**
- d) The height of the left subtree and the right subtree of every node differs by at least 1
- 19 **What is the time complexity of search, insert, and delete operations in a 2-3 tree?**
- a) $O(\log n)$
- c) $O(n \log n)$
- b) $O(n)$**
- d) $O(2^n)$
- 20 **What happens when a node in a 2-3 tree is split due to the insertion of a new key?**
- a) The node is replaced by two new nodes, each with one key
- c) The node is not split, and the new key is added to the existing node
- b) The node is replaced by three new nodes, each with one key**
- d) The node is replaced by two new nodes, each with one or two keys**
- 21 **In a max-heap, which of the following is true?**
- a) The parent node is smaller than its child nodes
- c) The tree is a binary search tree
- b) The parent node is greater than or equal to its child nodes**
- d) The elements are stored in ascending order
- 22 **What is the time complexity to build and heapify a heap of n elements?**
- a) $O(n \log n)$
- c) $O(n)$
- b) $O(\log n)$**
- d) $O(n^2)$
- 23 **You are given an array: [3, 9, 2, 1, 4, 8]. After building a max-heap from this array, what will be the root element?**
- a) 1
- c) 2
- b) 9**
- d) 8**
- 24 **Which of the following applications is a heap NOT suited for?**
- a) Implementing a priority queue
- c) Sorting data in ascending order
- b) Finding the median of a data stream**
- d) Depth-first search of a graph**
- 25 **Which of the following is NOT true about heap sort?**
- a) It is an in-place sorting algorithm
- c) It has a time complexity of $O(n \log n)$
- b) It is a stable sorting algorithm**
- d) It uses the heap data structure
- 26 **What makes Counting Sort a distribution-based algorithm?**
- a) It uses element frequencies to determine sorted positions.
- c) It recursively splits and merges data.
- b) It divides the array into smaller subarrays.**
- d) It compares each pair of elements to sort the array.**
- 27 **Which sorting algorithm explicitly trades off space for time by using extra space to count occurrences of elements?**
- a) Merge Sort
- c) **Sorting by counting**
- b) Bubble sort**
- d) Quick sort
- 28 **What is the time complexity of distribution counting sort?**
- a) $\Theta(n^2)$
- c) $O(n)$
- b) $\Theta(n)$**
- d) $O(n^2)$**
- 29 **What is the time complexity of comparison counting sort?**

- a) $\Theta(n^2)$
c) $O(n^2)$

b) $O(n)$
d) $\Theta(n)$

30 **Sorting by counting trades off space for time efficiency by:**
a) Using recursion to divide the array into smaller parts.
c) Avoiding comparisons between elements.

b) **Allocating extra memory for the frequency and cumulative arrays.**
d) Reducing the overall size of the input array.

31 **In Horspool's algorithm, when the last character of the pattern does not match the text, what action does the algorithm take?**
a) Shift the pattern by 1 character to the right
c) **Shift the pattern by the value stored for the text character in the shift table**

b) Perform a full re-check from the beginning of the pattern
d) Abort the search

32 **In Horspool's algorithm, when does the algorithm perform a full comparison of the pattern with the text?**
a) Every time a mismatch occurs
c) Only when the shift table suggests no shift

b) Only when a match is found
d) Initially, when the first character in the pattern is compared with the first character of the text

33 **In Horspool's algorithm, what is the primary purpose of the shift table?**
a) To store the indices of pattern matches
c) **To store the number of shifts required for each character in the pattern**

b) To store the positions of all characters in the pattern
d) To store the hash values of pattern characters

34 **In Horspool's algorithm, which table is used to store the shift values for the pattern matching process?**
a) **Shift Table**
c) Match Table

b) Hash Table
d) Character Table

35 **What is the primary advantage of Horspool's algorithm over the naive string matching algorithm?**
a) It guarantees finding all occurrences in linear time.
c) It always matches patterns in fewer comparisons than the naive algorithm.

b) **It uses a precomputed shift table to skip unnecessary comparisons.**
d) It works in constant space

36 **Which of the following is an example of a problem that can be solved using dynamic programming?**
a) Binary search
c) Merge sort

b) **Knapsack problem**
d) Depth First Search

37 **In dynamic programming, the solution to a problem is built up from:**

- a) A single recursive call
 - b) **A series of overlapping subproblems**
 - c) A greedy approach
 - d) A divide and conquer method
- 38 **When solving a problem using dynamic programming, what is the role of "optimal substructure"?**
- a) **It ensures that solving subproblems will always result in the correct solution for the entire problem.**
 - b) It guarantees that the problem can be solved in constant time.
 - c) It allows for efficient recursion without overlapping subproblems.
 - d) It simplifies the problem by eliminating unnecessary subproblems.
- 39 **What is the time complexity of solving the 0/1 knapsack problem using dynamic programming with an array-based approach?**
- a) $O(n^2)$
 - b) **$O(n * W)$ where n is the number of items and W is the maximum weight**
 - c) $O(W)$
 - d) $O(n)$
- 40 **What does the term "overlapping subproblems" refer to in dynamic programming?**
- a) Problems that can be solved using greedy algorithms
 - b) **Subproblems that appear multiple times in the recursive solution**
 - c) Subproblems that are completely independent
 - d) Subproblems that cannot be solved independently
- 41 **What is the primary purpose of the Floyd-Warshall algorithm?**
- a) Finding the shortest path between two specific nodes
 - b) **Finding the shortest paths between all pairs of nodes in a graph**
 - c) Finding the minimum spanning tree of a graph
 - d) Finding the longest path in a weighted graph
- 42 **Which type of graph is required for the Floyd-Warshall algorithm?**
- a) **Directed or undirected weighted graphs**
 - b) Only directed weighted graphs
 - c) Only undirected graphs
 - d) Only acyclic graphs
- 43 **How does the Floyd-Warshall algorithm handle negative weight cycles?**
- a) **It detects them and stops execution**
 - b) It ignores them and continues finding shortest paths
 - c) It gives incorrect results
 - d) It runs indefinitely
- 44 **Which technique is used to solve the 0/1 Knapsack problem efficiently?**
- a) Greedy algorithm
 - b) Divide and conquer
 - c) **Dynamic programming**
 - d) Depth-first search
- 45 **What does the Floyd-Warshall algorithm store in its intermediate matrix?**
- a) Edge weights only
 - b) **Shortest path distances**
 - c) Number of edges in the shortest path
 - d) The longest path in the graph
- 46 **Which of the following is a major limitation of the Floyd-Warshall algorithm?**
- a) Cannot handle negative weights
 - b) High space complexity
 - c) Cannot find shortest paths for undirected graphs
 - d) **Cannot find shortest paths for large graphs efficiently**
- 47 **What is the space complexity of the Floyd-Warshall algorithm?**

- a) $O(V)$
- b) $O(V^2)$
- c) $O(V^3)$
- d) $O(V \log V)$

48 **What is the Knapsack problem primarily used for?**

- a) Finding the shortest path in a graph.
- b) **Maximizing the value of items in a knapsack without exceeding its capacity.**

- c) Sorting a list of items.
- d) Finding the minimum spanning tree.

49 **Which of the following is true about memory function?**

- a) **It is a top-down approach**
- b) It is a bottom-up approach.
- c) It does not use recursion.
- d) It is only applicable to sorting algorithms.

50 **Which of the following problems can be solved using memory function?**

- a) **0/1 Knapsack algorithms**
- b) Travelling salesman problem
- c) Job assignment
- d) All of the above.

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Multiple Choice Questions

Unit-3

- 1` Which of the following is true?
a) **Prim's algorithm initialises with a vertex** b) Prim's algorithm initialises with a edge
c) Prim's algorithm initialises with a vertex which has smallest edge d) Prim's algorithm initialises with a forest
- 2 Prim's algorithm is a _____.
a) Divide and conquer algorithm b) **Greedy algorithm**
c) Dynamic Programming d) Approximation algorithm
- 3 Which of the following is false about Prim's algorithm?
a) It is a greedy algorithm b) **It constructs MST by selecting edges in increasing order of their weights**
c) It never accepts cycles in the MST d) It can be implemented using the Fibonacci heap
- 4 Kruskal's algorithm is used to _____.
a) **find minimum spanning tree** b) find single source shortest path
c) find all pair shortest path algorithm d) traverse the graph
- 5 What is the time complexity of Kruskal's algorithm?
a) $O(\log V)$ b) $O(E^2)$
c) **$O(E \log V)$** d) $O(V \log E)$
- 6 Which of the following is true?
a) Prim's algorithm can also be used for disconnected graphs b) Prim's algorithm is simpler than Kruskal's algorithm
c) In Kruskal's sort edges are added to MST in decreasing order of their weights d) **Kruskal's algorithm can also run on the disconnected graphs**
- 7 Which of the following algorithms is the best approach for solving Huffman codes?
a) exhaustive search b) **greedy algorithm**
c) brute force algorithm d) divide and conquer algorithm
- 8 Dijkstra's Algorithm is used to solve _____ problems.
a) All pair shortest path b) **Single source shortest path**
c) Network flow d) Sorting
- 9 Dijkstra's Algorithm is the prime example for _____.
a) **Greedy algorithm** b) Branch and bound
c) Back tracking d) Dynamic programming
- 10 In Huffman coding, data in a tree always occur?
a) roots b) left sub trees
c) **leaves** d) right sub trees
- 11` What is the primary advantage of the Bellman-Ford algorithm over Dijkstra's algorithm?
a) Works only for positive weights b) **Can handle negative weight edges**

- c) More efficient than Dijkstra's algorithm d) Uses a priority queue
- 12 What is the time complexity of the Bellman-Ford algorithm for a graph with V vertices and E edges?
- a) $O(V)$ b) $O(E)$
c) $O(V^2)$ **d) $O(VE)$**
- 13 If a graph contains a negative-weight cycle, what will Bellman-Ford algorithm do?
- a) Return shortest path normally **b) Terminate and output a warning**
c) Enter an infinite loop d) Ignore the cycle and compute shortest paths
- 14 What is the key principle behind edge relaxation in the Bellman-Ford algorithm?
- a) Updating distances to adjacent vertices if a shorter path is found** b) Assigning random weights to edges
c) Removing unnecessary edges d) Checking for cycles in the graph
- 15 In the state space tree used for backtracking, what do the nodes represent?
- a) The entire solution **b) Partial solutions leading to the final solution**
c) The weight of an edge d) The adjacency matrix
- 16 What is the purpose of the pruning step in backtracking algorithms?
- a) To reduce the number of recursive calls b) To guarantee an optimal solution
c) To avoid checking already visited nodes **d) To eliminate branches that cannot lead to a valid solution**
- 17 In the subset-sum problem, when do we backtrack?
- a) When the sum of the selected elements exceeds the target sum** b) When the sum of the selected elements is equal to the target sum
c) When all elements have been considered d) When a subset has been found
- 18 How does backtracking differ from brute force?
- a) Backtracking does not explore all possibilities **b) Backtracking avoids unnecessary computations by pruning**
c) Backtracking is faster in all cases d) Backtracking guarantees an optimal solution
- 19 In the subset-sum problem using backtracking, what is the worst-case time complexity?
- a) $O(N)$ **b) $O(2^N)$**
c) $O(N \log N)$ d) $O(N^2)$
- 20 What does the Bellman-Ford algorithm detect that Dijkstra's algorithm cannot?
- a) Shortest paths b) Graph cycles
c) Negative-weight cycles d) Directed edges
- 21 Which of the following is the primary purpose of the Branch and Bound method?
- a) To find the exact solution of a problem** **b) To eliminate suboptimal solutions and prune the search space**
c) To provide approximate solutions d) To reduce the problem into smaller sub-

only

problems

- 22 Which of the following is an example of a combinatorial optimization problem solved using Branch and Bound?
- a) Sorting numbers in ascending order
 - b) Multiplying two matrices
 - c) Searching for an element in a list
 - d) **Solving the traveling salesman problem (TSP)**
- 23 In the Job Assignment Problem, what does the objective function typically aim to minimize?
- a) The time to complete all tasks
 - b) The number of tasks assigned
 - c) **The total cost of assigning tasks to workers**
 - d) The total time worked by all workers
- 24 NP-Complete problems are:
- a) Solvable in polynomial time
 - b) Solvable using dynamic programming
 - c) **As hard as the hardest problems in NP**
 - d) Always solvable using Branch and Bound
- 25 In the Knapsack problem, if we are maximizing value and minimizing weight, which method would typically be applied?
- a) **Branch and Bound**
 - b) Greedy method
 - c) Dynamic programming
 - d) Backtracking
- 26 Which of the following statements is true about NP-Hard problems?
- a) They are solvable in polynomial time
 - b) They are more complex than NP-Complete problems
 - c) They cannot be solved in polynomial time, but their solutions can be verified in polynomial time
 - d) **They are at least as hard as NP-Complete problems**
- 27 Which problem is considered NP-Hard?
- a) **Traveling Salesman Problem (TSP)**
 - b) Shortest path problem
 - c) Knapsack problem
 - d) Sorting problem
- 28 Which of the following is NOT a common technique for solving the Job Assignment Problem?
- a) Linear Programming
 - b) **Greedy Algorithm**
 - c) Branch and Bound
 - d) Dynamic Programming
- 29 Which of the following problems is not NP-Complete?
- a) Knapsack Problem
 - b) Subset Sum Problem
 - c) Finding the longest path in a graph
 - d) **Sorting a list of integers**
- 30 Which of the following is the most efficient approach to solve a Knapsack problem with a large number of items?
- a) Brute force search
 - b) Greedy algorithm
 - c) **Dynamic programming or Branch and Bound**
 - d) Backtracking